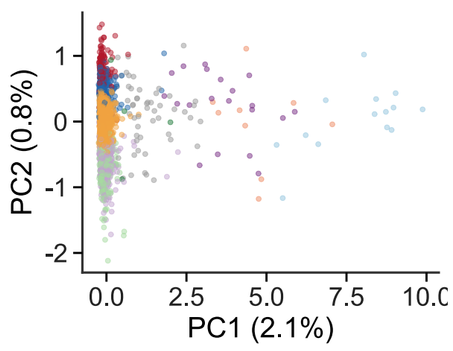
Figure 1 | Inverse optimal transport and current benchmark design

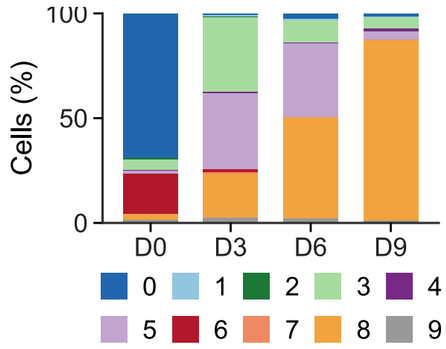
Data preparation

GSE228154 · 3,208 cells

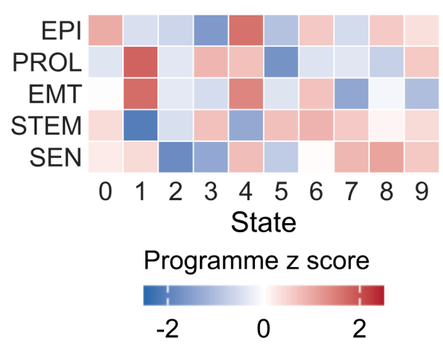
Expression PCA



Treatment-state composition



State × programme features



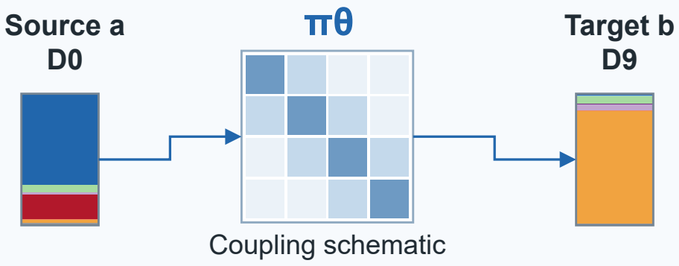
QC → state labels → a, b, φ

Context / lineage metadata
Independent prospective inputs

1 | Identify transport costs and test transfer

Fig. 2

From cell states to transport costs



Balanced OT
π1 = a; πᵀ1 = b

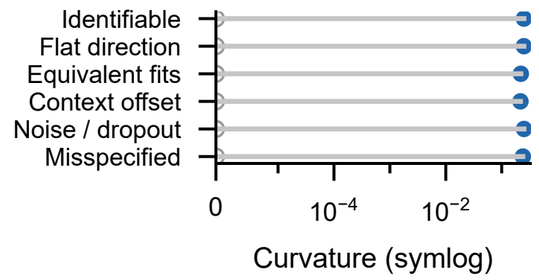
UOT-IOT
π1 = a; μ DKL(πᵀ1 // b)

$$C\theta(i,j) = C_0(i,j) - \theta^T\varphi(i,j)$$

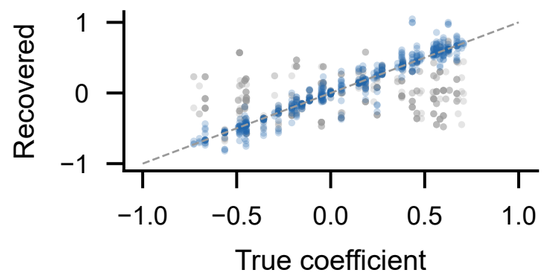
Inverse fit → θ and πθ

Curvature | Known truth | Restarts

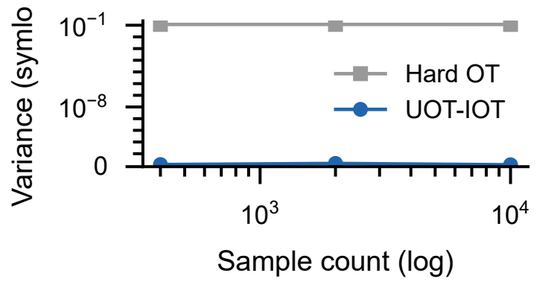
Known-truth curvature



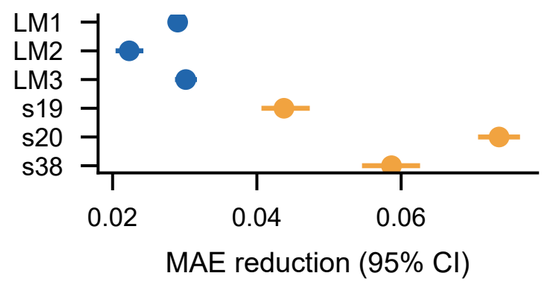
Synthetic coefficient recovery



Direction stability



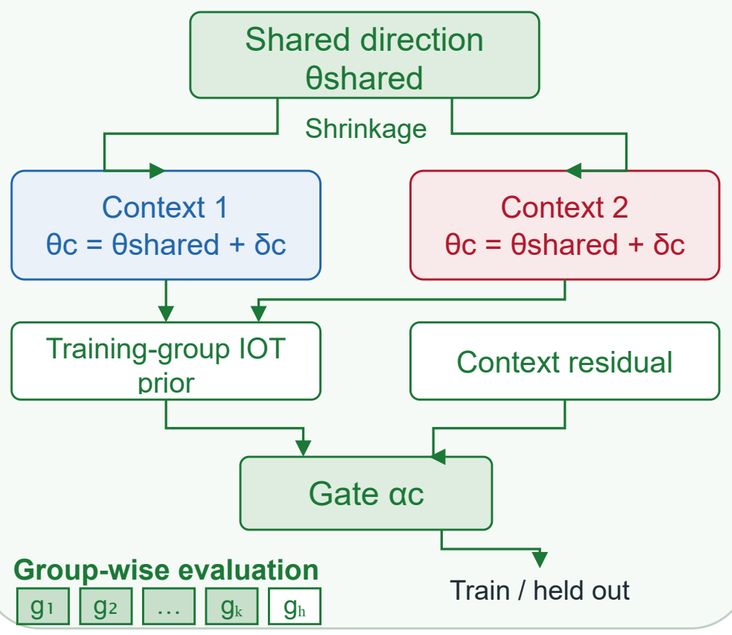
Frozen transfer · six sites



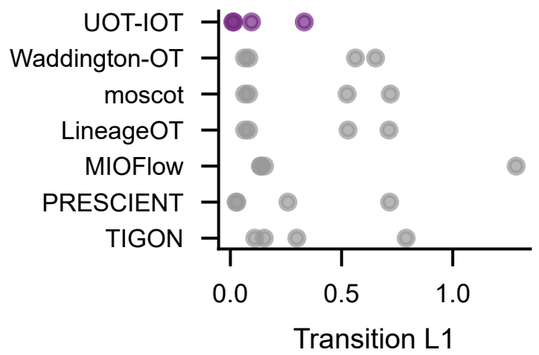
2 | Context model and common method comparisons

Figs. 3, 5

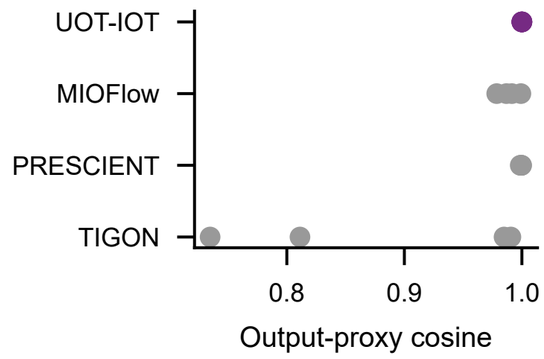
GH-IOT: shared and context effects



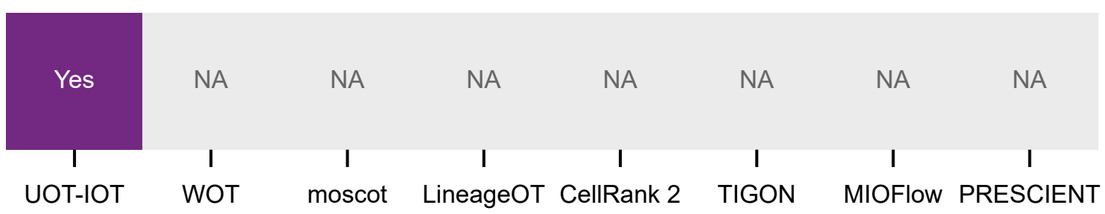
Lineage transition recovery



Output-proxy stability



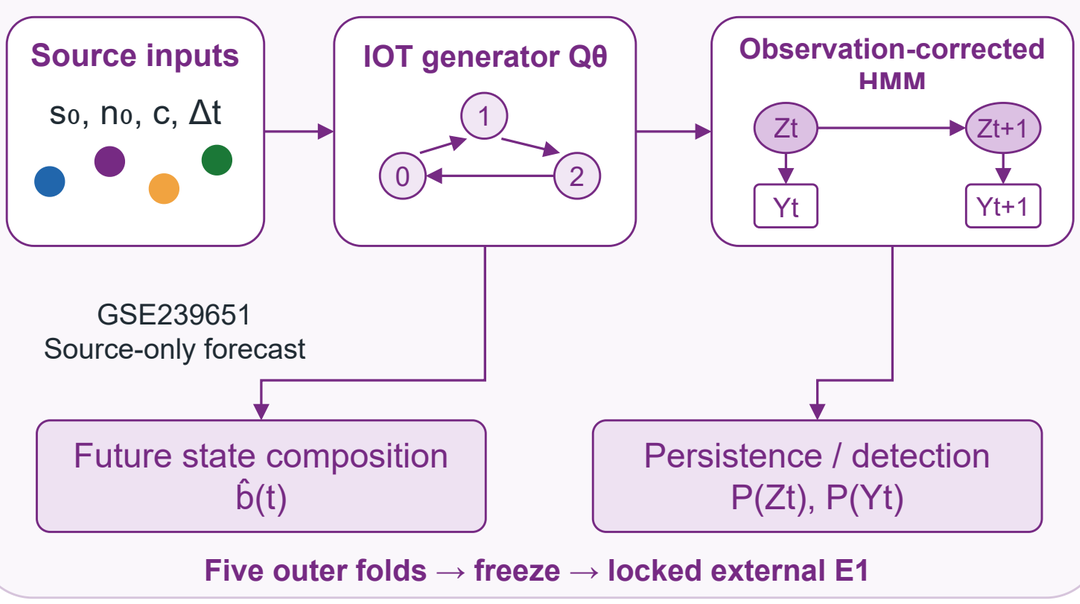
Comparable signed feature parameters · adapter capability



3 | Forecast future composition and evaluate calibration

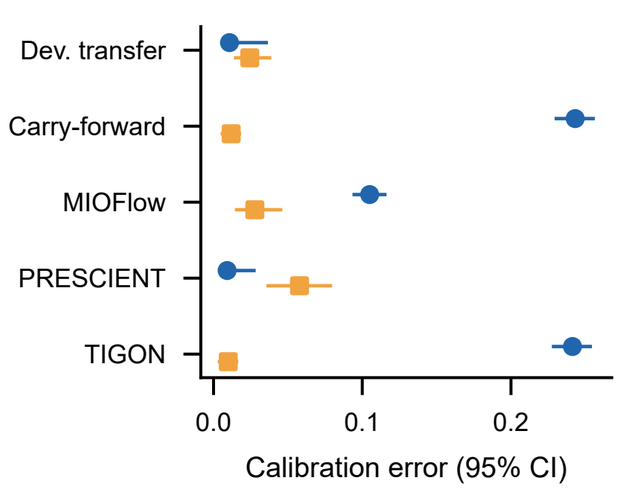
Fig. 4

PERSIST-IOT: source inputs and observation model



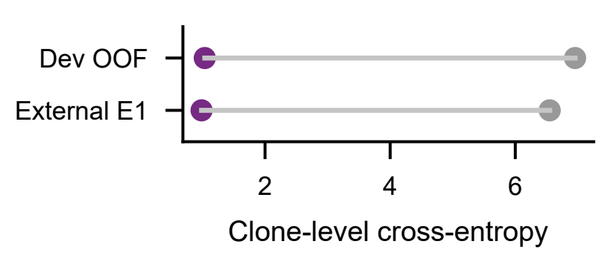
Five outer folds → freeze → locked external E1

Population calibration

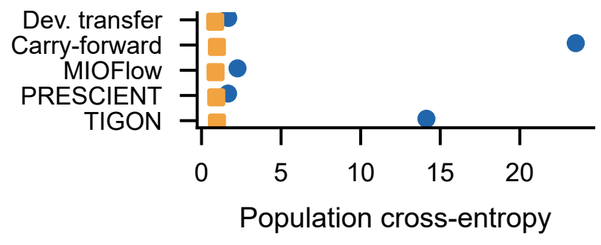


Blue: GSE140802 · Orange: GSE239651 expt2

Clone-level composition



Population-level composition



B1: six scenarios × three sample sizes × five seeds • B2: four lineage panels • B3: clone and population forecasts